WEEK 05 REPORT

MID-REVIEW – THE DATA AUDIT

MONEYBALL – IPL PLAYER VALUATION

24ADI204 – Data Science and Visualization

Data-Viz Hack-A-Bit (2026–2027) | Team Number: 05

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Presentation of the cleaned dataset and initial statistical findings.

1. Executive Summary

Week 05 is the Mid-Review milestone titled **The Data Audit** in the DSV Rule Book. The purpose is to present the cleaned dataset, explain the cleaning decisions and communicate initial statistical findings. This report consolidates the verified Week 03 cleaning results and Week 04 EDA findings for the review. Week 05 itself does not claim additional cleaning or transformation.

2. Problem Statement and Objectives

The IPL master dataset contains batting, bowling, player-profile and auction-related fields with missing values and extreme observations. The mid-review must demonstrate that the team understands the original data quality, can justify the cleaning choices and can communicate meaningful statistical findings.

Objectives:
• Present the cleaned IPL dataset and verify its dimensions and quality.
• Explain missing-value handling and duplicate checks.
• Summarize IQR and Z-score outlier detection and the decision to retain valid extremes.
• Present selected descriptive and EDA findings.
• Identify limitations and prepare for Week 06 feature engineering.

3. Dataset Profile

The project uses the merged IPL player-season dataset. The Week 03 cleaned file is named **IPL_Moneyball_Cleaned_Week03.csv**. The verified project profile is summarized below.

Property	Value
Rows	3,311
Columns	27
Season coverage	2008–2026 (19 seasons)
Distinct player values	794
Numerical / categorical columns	17 / 10
Missing cells after Week 03	13,099
Duplicate rows	0

The dataset contains player identifiers, season, batting and bowling statistics, player/team descriptors and auction-related fields. The original source URLs and complete source-data provenance were not established in the reviewed materials.

4. Cleaning Methodology and Decisions

Missing values: The original dataset contained 17,839 missing cells. Missing values in role (2,418), nationality (2,296) and bowl_style (26) were replaced with the category label *Unknown*. This handled 4,740 cells. The remaining 13,099 missing cells were retained where a defensible value could not be inferred. In particular, missing auction details and undefined performance averages were not filled with arbitrary zero or overall averages.

Duplicate check: The original dataset contained zero duplicate rows. Therefore, no records were removed.

Outlier review: Boxplots, IQR and Z-score were used to detect potential outliers. The counts below are column-level counts and must not be added as if they represented unique players.

Column	IQR outliers	Z-score outliers
runs	156	46
strike_rate	182	20
batting_avg	39	25
wickets	203	44
economy	107	31

Column	IQR outliers	Z-score outliers
final_price_lakh	130	Not calculated

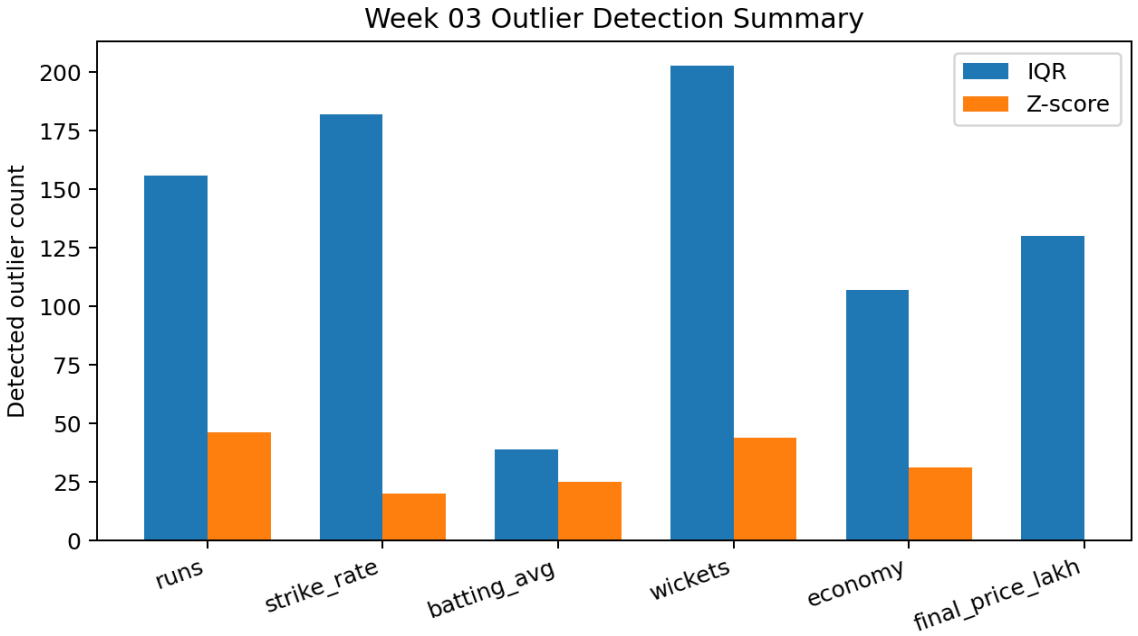


Figure 1. Outlier counts documented in the Week 03 cleaning report. Z-score was not calculated for final_price_lakh.

Detected extremes were not automatically deleted or capped. Very high runs, wickets, strike rates, economy values or auction prices can be genuine IPL observations. Without evidence of data-entry error, retaining them is more defensible.

5. Initial Statistical Findings from Week 04

Week 04 examined numerical distributions, categorical frequencies, pairwise relationships and correlations. The strongest verified positive correlations are listed below.

Variable pair	Pearson correlation
Runs vs Balls Faced	0.983936
Balls Bowled vs Runs Conceded	0.982674
Balls Bowled vs Matches Bowled	0.976987
Runs vs Fours	0.964513

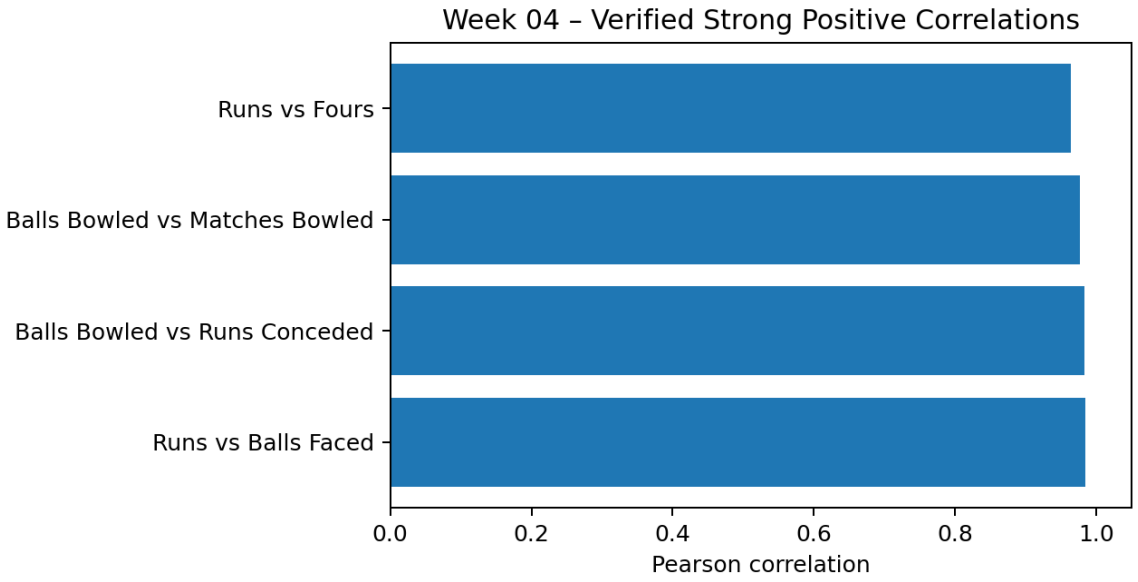


Figure 2. Strong positive correlations reported in Week 04.

These values indicate strong positive linear associations in the dataset. Runs and balls faced move together, as do balls bowled and runs conceded. Correlation does not prove causation.

6. Leading Players by Aggregate Runs

Rank	Player	Total runs
1	V Kohli	9,040
2	RG Sharma	7,183
3	S Dhawan	6,769

These are the top three values confirmed in the Week 04 analysis record. The report does not extend the ranking beyond verified results. Aggregate runs alone are not a complete measure of player value.

7. Mid-Review Discussion and Limitations

The cleaned dataset retains its original row and column counts because the selected missing-value treatment replaced categorical values rather than deleting records or columns. Remaining missingness is documented. Outlier counts indicate potential extremes, not necessarily errors. Price-related analysis may be limited by missing auction information. Player value should eventually consider batting, bowling, role, match context and auction information together.

8. Conclusion and Next Steps

The project has completed dataset inspection, targeted missing-value treatment, duplicate checking, outlier detection and initial EDA. The Week 05 review consolidates this evidence for presentation. The next scheduled activity, Week 06, is Feature Engineering: normalization, scaling, categorical encoding and skewness correction. Those transformations are not part of this Week 05 report.

9. Project Files and References

Cleaned dataset: IPL_Moneyball_Cleaned_Week03.csv
Week 05 review notebook: Week05_Mid_Review_Data_Audit_Team05.ipynb
Repository: https://github.com/Mithuinps/24ADI204-DSV-team-05/tree/main

References:

1. Data-Viz Hack-A-Bit (2026–2027), 24ADI204 – Data Science and Visualization, Design Competition Rule Book, Week 05: Mid-Review – The Data Audit.
2. Team Week 03 report: The Cleaning Sprint.
3. Team Week 04 EDA findings: EDA Deep Dive.
4. Team project repository.

Note: This report summarizes results already documented in Weeks 03 and 04. Run the Week 05 notebook with the actual cleaned CSV before presenting so the verification and descriptive-statistics outputs are generated from the file used by the team.